Reduction to Linear Form

Questions 1–32

Question 1. Answer sheets for questions marked with a (*) can be found at TLMaths.com Show that $y = ax^b$ may be written in the form

$$\log_{10} y = b \log_{10} x + \log_{10} a$$

Question 2. Show that $y = a \times b^x$ may be written in the form

$$\log_{10} y = x \log_{10} b + \log_{10} a$$

Question 3. Show that $y = ae^{bx}$ may be written in the form

$$\ln y = bx + \ln a$$

Question 4. A model for some data is given as

$$y = 2 \times 1.05^x$$

Show that the graph of $\log_{10} y$ against x will be a straight line.

Question 5. A model for some data is given as

$$y = 0.5x^{1.2}$$

Show that the graph of $\ln y$ against $\ln x$ will be a straight line.

Question 6. A model for some data is given as

$$y = 0.4 \times 0.88^{-x}$$

Show that the graph of $\log_{10} y$ against x will be a straight line.

Question 7. For each of the following, rearrange to find an equation of y in terms of x, giving your answer in the form $y = ax^b$ where a and b are values given to 2 significant figures:

a) $\log_{10} y = 0.478 + 5 \log_{10} x$
b) $\ln y = 3 \ln x + 0.815$
c) $\log_{10} y = -1.347 - 0.89 \log_{10} x$
d) $\ln y = 0.51 \ln x - 1.15$

Question 8. For each of the following, rearrange to find an equation of y in terms of x, giving your answer in the form $y = a \times b^x$ where a and b are values given to 2 significant figures:

a) $\log_{10} y = 0.214x - 1.54$
b) $\ln y = 0.555x + 0.444$
c) $\ln y = -0.235x + 0.159$
d) $\log_2 y = 0.895x + 2.03$

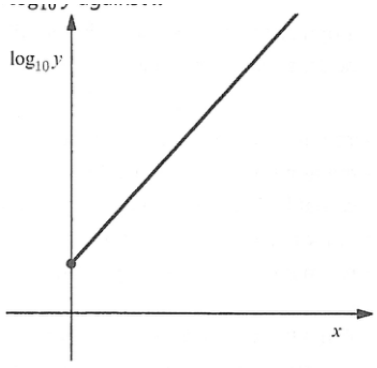
Question 9. Convert each of the following into the form $\ln y = a + bx$ where a and b are constants to be found:

a) $y = 40e^{-0.2x}$

b) $y = 0.5 \times 1.04^{0.5x}$

c) $y = 1.62 \times 2.13^{-0.2x}$

Question 10. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against x



A linear equation is found to best fit the data. In each of the following cases, find a model in the form

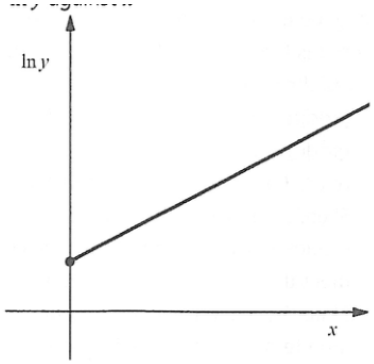
$$y = a \times b^x$$

where a and b are values to be found:

a) gradient of 3 and an intercept of 0.5

b) gradient of 0.25 and an intercept of 0.0714

Question 11. A data set (x, y) has been plotted on a graph of $\ln y$ against x



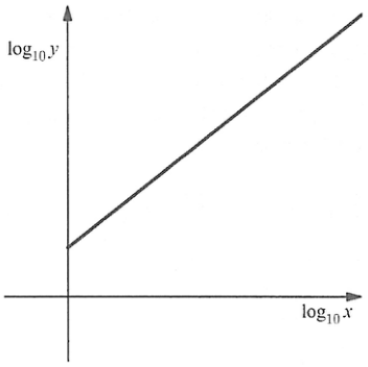
The linear equation found to best fit the data. In each of the following cases, find a model in the form

$$y = ae^{bx}$$

where a and b are values to be found:

a) gradient of 4 and an intercept of 2
b) gradient of 0.75 and an intercept of 0.1

Question 12. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against $\log_{10} x$



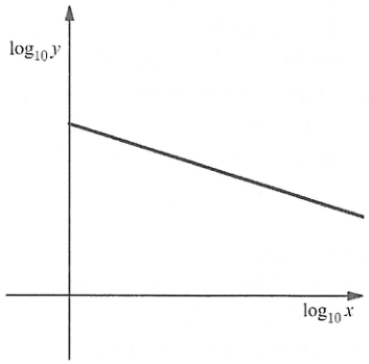
The linear equation found to best fit the data. In each of the following cases, find a model in the form

$$y = ax^n$$

where a and n are values to be found:

a) gradient of 4 and an intercept of 0.1
b) gradient of 0.5 and an intercept of 4

Question 13. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against $\log_{10} x$ The linear equation found to best fit the data.



In each of the following cases, find a model in the form

$$y = ax^n$$

where a and n are values to be found:

a) gradient of -5 and an intercept of 0.12
b) gradient of -0.4 and an intercept of 1.17

Question 14. A straight-line graph of $\ln y$ against $\ln x$ passes through the y-axis at 2 and the point with coordinates $(1.8, 2.9)$. Find the equation which represents the equation of the line in terms of y and x

Question 15. A straight-line graph of $\log_{10} y$ against $\log_{10} x$ passes through the y-axis at 3 and the point with coordinates $(1.6, 3.704)$. Find the equation which represents the equation of the line in terms of y and x

Question 16. A straight-line graph of $\log_{10} y$ against x passes through the y-axis at 0.5 and the point with coordinates $(1.01, 0.803)$. Find the equation which represents the equation of the line in terms of y and x

Question 17. A straight-line graph of $\ln y$ against x passes through the points $(1.1, 3.06)$ and $(2.4, 4.12)$. Find the equation which represents the equation of the line in terms of y and x

Question 18. (*) The number of visits to a website y since its launch x days before is recorded in the following table:

Days since launch, x	Number of visits, y
2	5
30	15
60	24
100	30
150	40

It is thought that the data may be modelled in the form

$$y = ax^b$$

a) Complete a table of values of $\log_{10} x$ and $\log_{10} y$, giving your answers to 3 decimal places.
b) Plot a graph of $\log_{10} y$ against $\log_{10} x$ and draw a line of best fit on your plot.
c) Find the y-intercept and the gradient of your line of best fit.
d) Show that $y = ax^b$ may be written in the form

$$\log_{10} y = \log_{10} a + b \log_{10} x$$

e) Use your graph to estimate the values of a and b to 1 decimal place.

Question 19. (*) The number of app downloads, N millions at time t months after launch may be modelled with the equation $N = at^b$. The data collected within the first 24 months is shown in the table below:

Time, t months	Downloads, N
3	12.1
5	24.7
8	50.3
10	65.9
18	170.2
24	249.1

a) Complete a table of values of $\log_{10} t$ and $\log_{10} N$, giving your answers to 3 decimal places.
b) Plot a graph of $\log_{10} N$ against $\log_{10} t$ and draw a line of best fit.
c) Use your graph to estimate the values of a and b to 1 decimal places.

Use your model to find

d) The amount of app downloads at 12 months.
e) The time it took to reach 200 million downloads.

Question 20. (*) The mass of radioactive iodine, m grams after t days can be modelled with the exponential equation $m = ae^{bt}$ where a and b are constants. Data is collected every 10 days and is recorded in the table below:

Time, t days	Mass, m grams
0	50
10	20.9
20	8.9
30	3.6
40	1.7

a) Complete a table of values of $\ln m$ and t, giving your answers to 3 decimal places.
b) Plot $\ln m$ against t and draw a line of best fit on your graph.
c) Find the values of a and b
d) Use your model to predict the mass of iodine after 50 days since the start.

Question 21. (*) Morgana has recorded how many minutes T she has spent exercising per week over the first 10 weeks of the year. She thinks the data may be modelled in the form $T = a \times t^b$ where t is the number of weeks since the beginning of the year. The data she collected is given in the table below:

Time, t weeks	Time spent exercising, T minutes
1	90
2	82
4	77
6	76
8	73
10	71

a) Plot a graph of $\log_{10} T$ against $\log_{10} t$ and draw a line of best fit on your graph.
b) Find the values of a and b and hence use your values of a and b to create a model in the form

$$T = a \times t^b$$

c) Interpret the meaning of the value a in the context of the model.
d) Use your model to find the amount of exercise she did 3 weeks after starting.
e) Comment on the validity of the model when $t = 0$
f) Comment on the long-term reliability of the model.

Question 22. Matt is trying to find a model to represent the price of an item every month. He thinks it will be modelled in the form $y = ax^b$ where y is the price and x is the time passed in months that has passed.
He plotted $\log_{10} y$ against x, and found the y-intercept to be 2.4 and calculated the gradient as 0.6
He is unsure what to do next and suspects he has gone wrong.

a) State the mistake in his method.
b) Working with the graph he plotted, write an equation of y in terms of x with the information you have been given.

Question 23. Dev has found a model for data that she has collected, but when she uses her model to make predictions the results are not what she expects.

x	y	$\ln x$
2	2300	7.74
4	2125	7.66
6	1975	7.59
8	1825	7.51
10	1675	7.24
12	1550	7.34

Gradient is -0.0395, intercept is 7.82

$$\ln y = -0.0395x + 7.82$$
$$y = 10^{-0.0395x+7.82}$$

a) Find the mistake in her working.
b) Find the equation of y in terms of x in the form

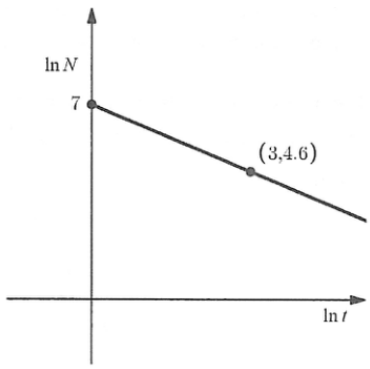
$$y = e^{px+q}$$

where p and q are constants to be found.

Question 24. The number of cinema admissions N millions in the UK, may be modelled by the equation

$$N = a \times t^b$$

where t is the number of years since 1954. The plot below shows the graph of $\ln y$ against $\ln t$, with two of points on the line labelled.

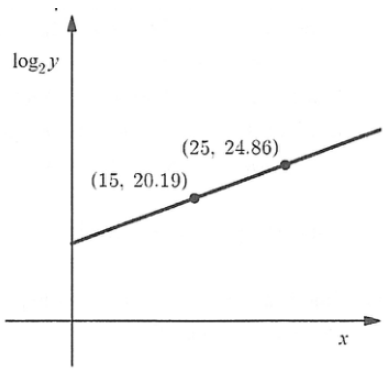


a) Find an equation for the line of best fit in terms of $\ln N$ and $\ln t$
b) Express N in terms of t
c) Use your model to predict the cinema admissions in 2020
d) Comment on the suitability of the model for predicting the number of cinema admissions in 2020

Question 25. The semi-conductor industry uses Moore's Law to predict the number of transistors y in a microprocessor, modelled by

$$y = a \times 2^{bx}$$

where x is the number of years since 1975. The plot below shows the graph of $\log_2 y$ against x, with two points on the line labelled.



a) Find an equation for the line of best fit in terms of $\log_2 y$ and x
b) Express y in terms of x
c) With reference to the model, explain the meaning of the value of a in context
d) Use your model to predict the number of transistors in 2030

Question 26. A content creator is monitoring the number of subscribers N to their channel, which may be modelled by

$$\log_{10} N = 0.238t + 1.975$$

where t is the number of subscribers t years after when the channel went live.

a) Show that $N = ab^t$ where a and b are constants to be found.
b) Interpret the meanings of the values a and b in the context of the model.
c) Use the model to calculate the total number of subscribers 10 years after the channel went live.

Question 27. Zain leaves a cup of hot water to cool, he collects his data, and establishes a model for the temperature $T°\text{C}$ over time t minutes

$$\ln T = -0.0701t + 4.585$$

a) Determine the initial temperature of the hot water

b) State an equation for T in the form $T = ae^{bt}$ where a and b are constants.

Question 28. It is found that the time it takes T days for a planet to orbit the sun may be given by

$$\ln T = 1.48 \ln d - 1.6$$

where d million kilometres is average distance of a planet from the Sun.

a) Find an equation for T in terms of d

Mars has an average distance of 228 million kilometres from the sun.

b) Using the model, find the number of days it will take Mars to orbit the sun, giving your answer to the nearest day.

It actually takes 687 days for Mars to orbit the sun.

c) Find the percentage error in the prediction made by the model.

Question 29. The planet Saturn has many moons. The radius of each of the moons is given by r million km. The number of days it takes a moon to complete one orbit of Saturn is given by T days. They may be modelled by

$$r = a \times T^b$$

	Radius, r million km	Period, T days
Titan	1.22	15.9
Rhea	0.53	4.5

a) Show that the equation may be written in the form $\ln r = \ln a + b \ln T$
b) Given the data in the table, find the values of a and b
c) Sketch the graph of r against T, labelling points which represent Titan and Rhea.

Another moon of Saturn, Dione has a period of 2.7 days and a radius of 3.8×10^5 kilometres.

d) Use your model to predict the radius of the moon giving your answer to 3 significant figures.
e) Find the % error of the prediction by the model.

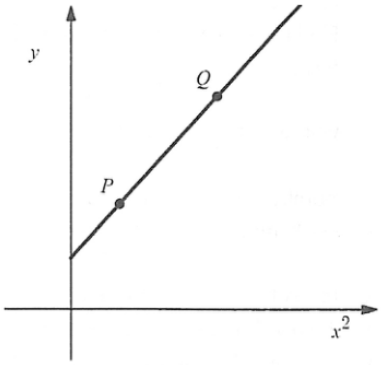
Question 30. The life expectancy of a person aged y years, may be defined to be a function of the number of years they have left to live, x years. A model for this may be given by

$$\ln y = \ln b - ax$$

It is known that when $x = 5, y = 71$ and when $x = 55, y = 24$

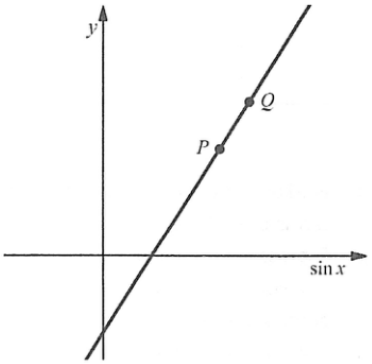
a) Find the value of a
b) Find the value of b
c) Use the model to find the life expectancy of someone aged 40
d) Find the age corresponding to a life expectancy of 25

Question 31. The diagram below shows a straight-line graph of y against x^2 passing through $P(0.3, 5.41)$ and $Q(4.7, 8.49)$



Find an expression for y in the form $ax^b + k$ where a, b and k are constants to be found.

Question 32. The diagram below shows a straight-line graph of y against $\sin x$ passing through $P(\frac{\pi}{2}, 5.3)$ and $Q(\frac{3\pi}{5}, 7.8)$



Write y in the form $a\sin(x) + b$ where a and b are constants to be found.